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Data engineering report: Group 10

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https://github.com/JasperStudentUA/DE-project

Introduction

Financial markets are shaped by a complex interplay of macroeconomic forces and investor sentiment. Among these, inflation stands out as one of the most consequential drivers of asset returns: it erodes purchasing power, influences central bank policy, and reshapes the relative attractiveness of different sectors of the economy. At the same time, the rise of digital platforms has introduced an entirely new dimension to market dynamics. The collective attention and anxiety of millions of retail investors, captured in real time through search engine query data. This report investigates the following research question: Can inflation data and public search interest predict sector ETF returns?

Exchange-Traded Funds (ETFs) that track specific economic sectors, such as energy, technology, healthcare, or consumer staples, offer a granular lens through which to study how macroeconomic conditions translate into investment outcomes. Unlike broad market indices, sector ETFs are particularly sensitive to inflation in heterogeneous ways: energy and commodities tend to benefit from rising prices, while rate-sensitive sectors such as real estate and utilities are typically hurt. This sectoral variation makes the question especially rich from an analytical standpoint. The inclusion of public search interest data, sourced from platforms such as Google Trends, is motivated by the growing body of behavioural finance literature suggesting that retail investor attention is a meaningful, and often leading, signal of market movements (Da et al., 2011). Search queries related to terms such as "inflation", "interest rates", or sector-specific keywords may reflect shifts in market sentiment before they manifest in prices, offering a potentially valuable predictive signal.

Data sources and collection

Three distinct data sources were integrated to construct the analytical dataset, each accessed programmatically through dedicated collection pipelines. All sources share a monthly temporal granularity and cover the period from January 2004 to April 2026, with 2004 chosen as the starting point to align with the earliest availability of Google Trends data.

Macroeconomic Data — Federal Reserve Economic Data (FRED)

Consumer Price Index (CPI) data was retrieved from the Federal Reserve Bank of St. Louis's FRED database via its public API. The specific series used was the seasonally adjusted CPI for all urban consumers (CPIAUCSL), which is the most widely referenced measure of price inflation in the United States. API credentials were managed securely through environment variables to avoid hardcoding sensitive information in source code.

From the raw CPI series, two derived variables were computed. First, year-over-year (YoY) inflation was calculated as the percentage change relative to the same month in the prior year, following the standard formula:

$$inflation_YoY_t = \frac{CPI_t - CPI_{t-12}}{CPI_{t-12}} \times 100$$

Second, each month was assigned a categorical inflation regime based on the computed YoY rate: *low* (below 2%), *moderate* (between 2% and 4%), or *high* (above 4%). These thresholds broadly reflect the US Federal Reserve's 2% inflation target and are consistent with classifications used in empirical macroeconomic literature. Observations with missing values arising from the twelve-month lag required for the YoY calculation were excluded prior to storage.

Financial Market Data — Yahoo Finance

Monthly sector ETF returns were sourced from Yahoo Finance, which provides free access to historical price data for a broad range of securities. Nine SPDR sector ETFs were selected, each tracking a distinct segment of the S&P 500:

- **Sector (Ticker):** Energy (XLE), Technology (XLK), Financials (XLF), Utilities (XLU), Healthcare (XLV), Consumer Discretionary (XLY), Consumer staples (XLP), Industrials (XLI), Materials (XLB)

Prices were collected at a monthly frequency starting from January 2004, with an adjustment applied to account for corporate actions such as stock splits and dividend distributions. This ensures that computed returns reflect the actual experience of a buy-and-hold investor, rather than being distorted by mechanical price changes unrelated to market performance. Monthly percentage returns were then computed as the first-order percentage change in adjusted closing prices:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100$$

Public Search Interest — Google Trends

Search interest data was retrieved from Google Trends, a publicly available platform that reports the relative popularity of search queries over time and geography. Thirteen search terms were collected, spanning both general macroeconomic sentiment and sector-specific investor attention:

- **Macro terms:** *inflation*, *recession*, *interest rates*, *oil price*

- **Sector terms:** *energy stocks*, *tech stocks*, *bank stocks*, *utility stocks*, *healthcare stocks*, *retail stocks*, *dividend stocks*, *manufacturing stocks*, *commodity stocks*

All queries were restricted to the United States and covered the full 2004–2026 timeframe. Google Trends returns a normalised search interest index scaled between 0 and 100, where 100 represents the peak search volume for a given term within the selected period. The inclusion of these variables is motivated by the behavioural finance literature, which suggests that retail investor attention, as proxied by search activity, can serve as a leading indicator of market movements (Da et al., 2011). Two important limitations of this data source merit acknowledgement. First, because each term is normalised independently rather than relative to a common baseline, values are not directly comparable across different search terms. Second, Google Trends does not expose a formal API, meaning data retrieval depends on an unofficial interface that is subject to rate limiting; conservative delays between requests were therefore introduced to ensure reliable collection.

Methodology

Data Storage

Raw data collected from each of the three sources was persisted in the Apache Parquet format, a columnar binary file format widely used in data engineering for its combination of efficient compression and fast analytical read performance. Storing intermediate outputs in Parquet rather than plain CSV ensures that data types, including dates and floating-point values, are preserved exactly across pipeline stages without requiring repeated parsing.

For structured querying and transformation, DuckDB was used as the analytical database engine. DuckDB is an embedded, in-process SQL database optimised for analytical workloads, offering the expressive power of SQL without the operational overhead of a client-server architecture. It is capable of reading Parquet files directly as external tables, which allowed the three raw data sources to be joined and transformed without first loading them into memory. Query results were persisted in a local DuckDB database file (analytics.db), providing a lightweight but durable store for intermediate and final tables throughout the pipeline.

This combination of Parquet for file-based storage and DuckDB for in-process analytical querying reflects a modern, lightweight approach to the data lakehouse pattern, where raw data is retained in open file formats and transformations are applied on demand through a query engine rather than a traditional database server.

Data Processing and Feature Engineering

The three raw Parquet files were integrated into a single master dataset through an inner join on the shared date column, retaining only months for which all three data sources held an observation. This join was performed entirely in SQL within DuckDB, producing an intermediate table containing all raw variables from each source. Feature engineering was subsequently applied within the same SQL environment using window functions, avoiding the need to export data to a separate processing framework for this step. A core design principle guiding all feature construction was the strict avoidance of look-ahead bias: all predictor variables were constructed such that they reflect information that would have been available to an investor at the start of the month whose returns are being predicted, not information from that month itself.

Specifically, the following features were derived from the raw variables:

- **Lagged inflation**: Year-over-year inflation with lags of one and three months, capturing the most recent observed inflation rate as well as its recent trajectory.

- **Rolling average inflation**: A three-month rolling mean of year-over-year inflation, intended to smooth short-term volatility and capture the prevailing inflation trend.

- **Lagged Google Trends**: All thirteen search interest series were lagged by one month, so that the model uses search activity from the prior month as a predictor of the following month's returns.

In addition, sector-specific features were constructed for each model individually at the analysis stage: a one-month lagged return and a three-month rolling average return, both capturing recent momentum in the sector's own price history. A month-of-year indicator was also added to account for known seasonal patterns in equity returns.

The fully processed master dataset was exported as a Parquet file and made available as input to the modelling pipeline.

Modelling and Analysis

A separate predictive model was trained for each of the nine sectors, following an identical methodology to ensure comparability of results across sectors. The target variable in each case was the return of the sector ETF in the following month, framing the task as a one-step-ahead regression problem.

XGBoost (Extreme Gradient Boosting) was selected as the modelling algorithm. XGBoost is an ensemble method based on gradient-boosted decision trees, well suited to tabular data with moderate dimensionality and known

for its robustness to outliers and non-linear feature interactions. Given the relatively modest size of the dataset, approximately twenty years of monthly observations, a shallow tree configuration was used to limit overfitting, with a maximum depth of three, a learning rate of 0.05, and 100 boosting rounds.

Model evaluation followed a temporal train-test split, with the first 80% of observations used for training and the remaining 20% reserved for out-of-sample testing. To assess generalisation more robustly during development, a walk-forward cross-validation procedure was also employed across three expanding folds, which respects the temporal ordering of the data and avoids leakage from future observations into training. Model performance was measured using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), and compared against a naive baseline, the three-month rolling average return, to assess whether the model adds predictive value beyond a simple heuristic.

Additional Data Engineering Tools

Beyond the core pipeline described above, two additional tools warrant mention. Apache Spark was incorporated at the model ingestion stage to read and validate the processed master dataset before handing it off to the XGBoost training routine. While the dataset in this project does not approach the scale at which Spark's distributed processing capabilities become strictly necessary, its inclusion demonstrates familiarity with production-grade data engineering tooling and would allow the pipeline to scale gracefully to larger datasets without architectural changes. All pipeline stages, being data collection, storage, processing, modelling, and visualisation, were implemented in Python, with dependency management and environment configuration handled through standard tooling. Credentials for external APIs were managed through environment variables rather than being embedded in source code, reflecting basic but important principles of secure pipeline design.

Analysis

Results

Six of nine sectors outperform the baseline. Healthcare, Industrials, and Consumer Discretionary see the largest gains, consistent with their sensitivity to macroeconomic cycle positioning. Consumer staples, Utilities, and Financials underperform, as their primary return drivers, commodity supply shocks, interest rate dynamics, and credit conditions, are not well captured by the current feature set.

SECTOR	MAE	MAE VS. BASELINE	DIRECTIONAL ACC.
CONSUMER STAPLES	3.58	-0.5%	52%
HEALTHCARE	4.08	+9.3%	52%
UTILITIES	4.37	-6.2%	52%
INDUSTRIALS	4.64	+9.5%	56%
MATERIALS	5.16	+5.5%	56%
TECHNOLOGY	5.26	+6.5%	58%
FINANCIALS	5.27	-6.9%	38%
ENERGY	6.12	+3.8%	54%
CONSUMER DISCRETIONARY	5.22	+11.7%	62%

As shown in Figure 1 The predicted series are consistently smoother than realized returns across all sectors, a structural consequence of mean-targeting ensemble regressors that capture conditional directional trends but systematically underestimate the magnitude of large monthly moves. This compression is most pronounced in Energy, which exhibits the sharpest divergence between actual and predicted values, while Technology and

Financials track medium-term momentum most coherently. The degree to which the model adds value over the naïve baseline is strongly sector-dependent is given in Figure 2, with improvements ranging from −6.9% (Financials) to +11.7% (Consumer Discretionary); sectors where sentiment and inflation proxies align closely with underlying return drivers benefit most, whereas rate- and commodity-driven sectors do not. This heterogeneity is reflected in the feature importance structure (Figure 3): the 3-month smoothed inflation average is the single strongest predictor overall, followed by the 1-month and 3-month lagged sector returns, while Google Trends features are collectively the most prevalent signal class in the top twelve — with *Search: oil price* ranking as the strongest individual Trends predictor. The narrow range of importance scores (0.048–0.079) indicates that no single feature dominates, suggesting the model draws from a distributed set of signals rather than relying on any one variable.

Conclusion

The models provide meaningful incremental predictive value for most sectors, with Google Trends emerging as a broadly useful non-traditional signal. Directional accuracy of 50–60% and systematic return compression limit practical trading utility in the current form. Extending the feature set to include yield-curve and commodity price data would most directly address underperforming sectors.

Reflection

The most significant practical challenge was collaborative version control. Setting up and working with GitHub as a team introduced friction that was not always anticipated: managing branches, resolving merge conflicts, and coordinating contributions across team members proved at times to be cumbersome. This experience underscores that process discipline is as important in data engineering projects as technical proficiency. AI coding assistants were used extensively throughout the project, specifically ChatGPT Codex, Perplexity, and Claude. These tools proved effective for accelerating development, generating boilerplate code, debugging pipeline errors, and suggesting implementation approaches for unfamiliar libraries. Beyond code generation, AI assistants were also used to support the writing process — helping to rephrase, structure, and refine sections of this report — as well as to generate diagnostic print statements and formatted outputs for intermediate data validation steps within the Python scripts. Rather than replacing understanding, these tools functioned as interactive reference tools that reduced time spent on routine coding and writing tasks. Correctness was ensured by critically reviewing all generated code and text prior to use, and by validating pipeline outputs against expected results at each stage of the project.

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